Research Thematic Areas 2026 2027

Advanced Topics in Artificial Intelligence

Choosing a research project

These thematic areas are research umbrellas, not predefined projects. Select an area, read relevant starting papers and independently develop a focused research question and plan. Projects may develop a method, compare existing approaches, create a dataset or benchmark, analyze model behavior, or evaluate a system in a specific setting.

After four weeks, each student submits a proposal written independently. It should set out the research question, motivation, relevant literature, proposed methodology, data and resources, evaluation plan, and a feasible schedule. Supervisors then interview students and assess the depth of their preparation and the suitability and rigor of the proposal and plan. Students use the interview feedback to refine the project and continue toward the research paper and poster.

A suitable proposal has a question narrow enough to investigate in one semester, a meaningful baseline and a way to interpret the results. The descriptions and research directions below illustrate open questions; they do not prescribe the method, scale or expected outcome of a student's project. Themes overlap: reasoning, robustness, explainability, uncertainty, adaptation, knowledge and evaluation can be studied across several modalities.

Learning Architectures and Representation

Graph Machine Learning

Scope

Graph machine learning studies data whose relationships matter as much as individual entities, from node and edge prediction to whole-graph tasks. Methods range from message passing and expressive GNNs to graph transformers, geometric and hypergraph models, and architectures without conventional convolution. Open questions include oversmoothing, oversquashing, scalability and generalization beyond the training distribution. Explainability is a further challenge: post-hoc, counterfactual and prototype-based explanations need careful evaluation, since a high fidelity score alone may mislead. Projects may connect graph methods to vision, biochemistry or recommendation, or address reproducibility and the lack of ground-truth explanation benchmarks.

Possible research directions

- Compare architectures under distribution shift or limited training data.
- Test whether a graph explanation remains faithful under perturbation.
- Design a reproducible benchmark that exposes a meaningful limitation in graph explanation or cross-domain transfer.
- Apply graph methods to images, biochemistry or recommendations.

Starting references

- Ying et al. (2019), GNNExplainer. Paper
- Zhang et al. (2021), ProtGNN. Paper
- Li et al. (2019), Graph Matching Networks. Paper
- Han et al. (2022), Vision GNN. Paper

Graph Language Models and Structured Reasoning

Scope

This area combines the structural processing of graph neural networks with the semantic knowledge and flexible language interface of large language models. Text representations of graphs can struggle with permutation invariance, limited context and multi-hop algorithmic tasks, while GNNs may lack the zero-shot flexibility and world knowledge of language models. Research explores graph serialization and tokenization, soft prompts, representation alignment, hybrid architectures and native graph-language models. A central question is whether a system uses graph structure or exploits textual patterns and benchmark artifacts. Careful comparisons can isolate the contributions of each modality and test transfer to chemistry, knowledge graphs, code analysis or planning.

Possible research directions

- Build an evaluation set that limits memorization or data contamination.
- Measure the contributions of graph structure and textual descriptions through ablations.
- Compare graph tokenization or hybrid graph-language designs.
- Study an application in chemistry, code, knowledge graphs or planning.

Starting references

- Petkar et al. (2025), A Graph Talks, But Who's Listening? Paper
- Fatemi et al. (2023), Talk like a Graph. Paper
- Zhang et al. (2024), Can LLM Graph Reasoning Generalize beyond Pattern Memorization? Paper
- Wang et al. (2024), LLMs as Zero-shot Graph Learners. Paper
- Wang et al. (2023), Can Language Models Solve Graph Problems in Natural Language? (NeurIPS). Paper

Training Adaptation and Alignment

Scope

Continual learning asks how a model can acquire new tasks or adapt to changing data without catastrophic forgetting or negative transfer. The broader area also includes reinforcement learning and preference-based alignment. In vision, language and multimodal systems, possible approaches include prompt tuning, parameter-efficient adapters, modular parameter isolation and dynamic routing. These choices matter when retaining training examples is restricted by privacy or computation. Multimodal work can use aligned vision-language features or structured auxiliary data, such as patient context, to guide mixture-of-experts routing and reduce spurious shortcuts or uncertainty outside the training distribution. Projects may examine stability against adaptation, task sequences, domain shifts and practical deployment on edge or clinical systems.

Possible research directions

- Compare adapter or prompt approaches under a realistic task sequence.
- Measure forgetting, transfer and resource cost together.
- Investigate routing or auxiliary metadata in a multimodal setting.
- Study how preference-based adaptation affects a clearly defined behavior, using appropriate baselines.

Starting references

- Wang et al. (2024), A Comprehensive Survey of Continual Learning. Paper
- De Lange et al. (2022), A Continual Learning Survey. Paper

- Wang et al. (2022), Learning to Prompt for Continual Learning. Paper
- Razdaibiedina et al. (2023), Progressive Prompts. Paper
- Ni et al. (2023), Continual Vision-Language Representation Learning with Off-Diagonal Information (ICML). Paper

Generative and Foundation Models

Generative AI for Images

Scope

Image generation has progressed from generative adversarial networks to diffusion and related methods. One research strand studies architecture: U-Net backbones, transformers and state-space models offer different trade-offs in quality, model size and computational cost. Another studies the training and sampling objective used to connect distributions, including diffusion, flow matching and newer transition-matching approaches. These choices affect both image quality and the speed of training or generation. Projects can investigate efficiency through architecture, distillation or sampling, or test a newer method on a defined image task with controlled comparisons. A feasible proposal should make its computational budget and evaluation criteria explicit.

Possible research directions

- Compare diffusion and flow-based methods at a fixed compute budget.
- Investigate whether an architectural change or distillation strategy improves efficiency without compromising quality.
- Evaluate sampling speed against quality for a specific image task.
- Explore a newer matching objective with a reproducible baseline.

Starting references

- Ho et al. (2020), Denoising Diffusion Probabilistic Models. Paper
- Peebles and Xie (2023), Scalable Diffusion Models with Transformers. Paper
- Lipman et al. (2022), Flow Matching for Generative Modeling. Paper
- Ma et al. (2024), SiT. Paper
- Shaul et al. (2025), Transition Matching (NeurIPS). Paper
- Su et al. (2023), Dual Diffusion Implicit Bridges for Image-to-Image Translation (ICLR). Paper

Generative AI for Language

Scope

This area studies language generation and translation across languages and cultures, as well as reasoning in large language models. Questions include factual reliability, bias, uncertainty, explainability, long-context behavior and robustness when instructions or evidence conflict with prior knowledge. The source also highlights linguistic priors in multimodal understanding and strategic, game-theoretic tasks involving collaborating or competing language-model agents. Research can assess how a model reaches an answer, whether it is grounded in available evidence, and how performance varies across languages and contexts. A proposal should define the task and failure mode, appropriate baselines and an evaluation that can distinguish genuine reasoning from plausible output.

Possible research directions

- Evaluate robustness of translation or generation across languages and cultural settings.
- Test factual grounding or uncertainty when evidence conflicts with prior knowledge.
- Study bias in recommendations or strategic reasoning.
- Investigate response clarity or long-document question answering.

Starting references

- Georgousis et al. (2026), Evaluating Counterfactual Strategic Reasoning in Large Language Models. Paper
- Stringli et al. (2025), Pitfalls of Scale. Paper
- Filandrianos et al. (2025), Bias Beware. Paper
- Thomas et al., I Never Said That. Paper
- Raptopoulos et al. (2025), PAKTON. Paper

Multimodal and Vision Language Models

Scope

Vision-language models combine visual and textual information, but good answers can reflect visual reasoning, learned shortcuts, linguistic priors or memorized knowledge. This area studies visual and multilingual understanding, counterfactual reasoning, explainability and robustness when an image conflicts with a model's prior knowledge. It also examines hallucination and the risks of relying on weak visual grounding in agents or other systems that act on multimodal outputs. Projects can use controlled changes to images, questions or languages to identify which evidence the model uses and where its reasoning fails.

Possible research directions

- Create controlled visual puzzles or counterfactual image-text pairs.
- Compare performance across languages while keeping visual content fixed.
- Test how a model resolves conflict between an image and its prior knowledge.
- Measure hallucination, confidence or explanation quality under image changes.

Starting references

- Lymperaiou et al., Reasoning or Pattern Matching? Probing Large Vision-Language Models with Visual Puzzles. Paper
- Li et al. (2023), Evaluating Object Hallucination in Large Vision-Language Models (EMNLP). Paper
- Liu et al. (2021), Visually Grounded Reasoning across Languages and Cultures (EMNLP; MaRVL). Paper
- Zhang et al. (2023), What If the TV Was Off? Examining Counterfactual Reasoning Abilities of Vision-Language Models (ICCV Workshop). Paper

Knowledge Reasoning and Trustworthy AI

Knowledge Augmented AI

Scope

Language models can draw on facts encoded during training or retrieve information from external sources, including text and other modalities. This area asks how factual associations are represented, where they are stored and how they can be updated. Probing and causal tracing can test whether particular internal components contribute to factual recall; targeted editing methods such as ROME and MEMIT aim to change selected associations without full retraining. Retrieval-augmented approaches offer another way to ground answers in accessible evidence. Projects should assess factual accuracy together with paraphrase generalization, locality and ripple effects on related knowledge, or compare stored knowledge with retrieved evidence.

Possible research directions

- Compare retrieval-augmented answers with answers from model parameters alone.
- Test factual editing for efficacy, paraphrase generalization and locality.
- Probe how a model represents a selected relation or fact.
- Evaluate multimodal retrieval on a defined question-answering task.

Starting references

- Petroni et al. (2019), Language Models as Knowledge Bases? Paper
- Geva et al. (2021), Transformer Feed-Forward Layers Are Key-Value Memories. Paper
- Meng et al. (2022), Locating and Editing Factual Associations in GPT. Paper
- Meng et al. (2023), Mass-Editing Memory in a Transformer. Paper
- Yao et al. (2023), Editing Large Language Models. Paper

Explainable and Interpretable AI

Scope

Explainability studies whether model behavior can be understood and assessed by people. Intrinsically interpretable approaches include depth-constrained optimal decision trees and sparse generalized linear or additive models, which balance accuracy with transparency. The Rashomon effect asks what can be learned from multiple comparably accurate but different models. For black-box systems, feature attribution, local surrogates and concept-based methods explain predictions after training, but their fidelity and stability must be tested. Mechanistic interpretability investigates internal representations, attention heads and circuits to trace how information produces specific behavior in a language model. Projects should identify the intended audience and what would count as a useful, faithful explanation.

Possible research directions

- Compare optimal and conventional decision trees under accuracy and depth constraints.
- Study multiple comparably accurate but structurally different models.
- Test the stability and fidelity of post-hoc explanations. Paper
- Trace a narrowly defined behavior through model components.

Starting references

- Hu et al. (2019), Optimal Sparse Decision Trees. Paper
- Rudin et al. (2022), Interpretable Machine Learning. Paper

- Semenova et al. (2023), A Path to Simpler Models Starts With Noise (NeurIPS). Paper
- Madsen et al. (2022), Post-hoc Interpretability for Neural NLP: A Survey (ACM Computing Surveys). Paper
- Wang et al. (2023), Interpretability in the Wild. Paper

Evaluation of AI Systems

Supervisors Lefteris, Maria L

Scope

AI evaluation must reflect both technical performance and human interaction in the setting where a system is used. Accuracy alone can miss reliability, safety, bias, usability, interaction quality and broader impact. This area develops context-sensitive criteria and methods for systems used in scientific and everyday work, medical decision support, culturally aware translation and human-AI creative activity, including music co-creation. Projects may diagnose limitations of established metrics or compare automatic judgments with expert and user assessments. Tasks can span language models for translation or semantic similarity and vision-language systems for image captioning or visual question answering, especially on challenging cultural data.

Possible research directions

- Audit a metric on culturally challenging translation or multimodal examples.
- Design a human evaluation protocol for a specific AI-assisted task.
- Compare technical and interaction-based measures of the same system.
- Analyze agreement, disagreement and bias across evaluators or metrics.

Starting references

- Fragiadakis et al. (2024), Evaluating Human-AI Collaboration. Paper
- Raees et al. (2024), From Explainable to Interactive AI. Paper
- Yao et al. (2024), Benchmarking Machine Translation with Cultural Awareness (Findings of EMNLP). Paper
- Kocmi et al. (2024), Error Span Annotation: A Balanced Approach for Human Evaluation of Machine Translation (WMT). Paper
- Liu et al. (2023), G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment (EMNLP). Paper

AI in Visual and Creative Domains

Vision 3D and Graph Based Vision

Scope

This area studies how AI systems represent and reason about the objects, geometry and relationships in visual scenes. In 3D vision, point clouds and other spatial representations support tasks such as object recognition, segmentation and detection. Graph-based methods model relations among points, regions or objects, allowing a system to use both local geometry and the wider scene context. A project might compare graph construction strategies, examine whether learned relationships improve performance over simpler point-based or image-based models, or test how well a method handles incomplete and noisy visual data. Students should define one visual task and dataset, identify a suitable non-graph baseline and evaluate both predictive performance and computational cost.

Possible research directions

- Compare a graph-based point-cloud model with a point-based baseline on 3D classification or segmentation.
- Investigate how neighborhood size or graph construction affects learned 3D features.
- Test robustness to sparse, noisy or partially occluded point clouds.
- Model object relationships in images as scene graphs and evaluate relation prediction.

Starting references

- Han et al. (2022), Vision GNN [also listed under Graph Machine Learning]. Paper
- Qi et al. (2017), PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation (CVPR). Paper
- Wang et al. (2019), Dynamic Graph CNN for Learning on Point Clouds (TOG). Paper
- Zellers et al. (2018), Neural Motifs: Scene Graph Parsing With Global Context (CVPR). Paper
- Shi and Rajkumar (2020), Point-GNN: Graph Neural Network for 3D Object Detection in a Point Cloud (CVPR). Paper

AI for Music

Scope

Music information retrieval brings together signal processing, machine learning, music theory and cognition to represent, analyze, retrieve and generate music from audio, symbolic notation and metadata. Watermarking can support attribution and provenance, while fingerprinting identifies recordings; similarity models examine relationships in rhythm, melody, harmony or timbre. Other directions include generation, symbolic-music ontologies that support analysis and reasoning, and high-quality annotated datasets. These datasets can address emotion and expressivity, performance technique, genre and style, or particular musical features. Projects may connect these strands, such as assessing music generation with human preferences, comparing similarity representations across cultures or examining whether identification methods remain robust to transformations made by generative models.

Possible research directions

- Test fingerprinting or watermarking robustness under transformations.
- Compare music similarity measures with human judgments.
- Develop an annotated dataset with a clear research purpose, documentation and an evaluation protocol.
- Evaluate music generation or a symbolic-music ontology.

Starting references

- Salah et al. (2025), Survey of Imperceptible and Robust Digital Audio Watermarking Systems. Paper
- Cano et al. (2005), A Review of Audio Fingerprinting. Paper
- Papaioannou et al. (2026), Cross-Cultural Music Similarity. Paper
- Grötschla et al. (2025), Benchmarking Music Generation Models and Metrics. Paper
- Kantarelis et al. (2023), Functional Harmony Ontology. Paper